

RESEARCH INTERESTS

Multimodal deep learning — Computer vision — vision-language models — LLM-based representation learning — generative modeling for augmentation and prediction — large-scale geospatial datasets — segmentation, detection, and forecasting.

EDUCATION

University of Virginia (UVA)

PhD in Computer Science; CGPA 3.97/4.00

Advisor: Madhav Marathe (Working on Multimodal AI and Vision-Language (VLM))

Virginia, USA

August 2022 - May 2027

Bangladesh University of Engineering and Technology (BUET)

B.Sc. in Computer Science; CGPA 3.86/4.00

Advisor: Eunus Ali (Worked on Graph Neural Network (GNN))

Dhaka, Bangladesh

February 2015 - May 2019

EXPERIENCE

Biocomplexity Institution, UVA

Graduate Research Assistant

Chalottesville, VA

November 2022 - Present

- Conducting research at the intersection of deep learning, visionlanguage modeling, and geospatial AI. Focused on multimodal image detection, segmentation, and generation using large-scale remote sensing data. (Published worked at **NeurIPS**, **KDD**, **IJCAI**, **WACV**)

DISA Lab

Research Assistant

University of Calgary, Canada

December 2020 - August 2021

- Conducted extensive natural language processing research leveraging transformer-based language models such as BERT, GPT, and their variants for textual classification tasks. (Published work at **IST** journal)

Ahsanullah University of Science and Technology

Lecturer

Dhaka, Bangladesh

July 2019 - August 2022

- Led and mentored undergraduate research on deep learning and NLP, using transformer-based LLMs like BERT and GPT for text detection, classification, and translation.

SELECTED PUBLICATIONS

N. C. Mandal*, O. B. Hoque*, M. Wilson, S. Swarup, S. K. Nouwakpo, A. Adiga, M. Marathe. **IRRISIGHT: A Large-Scale Multimodal Dataset and Scalable Pipeline to Address Irrigation and Water Management in Agriculture**, *Neural Information Processing Systems (NeurIPS 2025)*.

- Built a multimodal benchmark combining satellite, crop, and soil data for irrigation analysis.
- Developed a scalable pipeline with VLMs, ViTs, and synthetic data generation tools.

N. C. Mandal*, O. B. Hoque*, A. Adiga, S. Swarup, M. Wilson, L. Feng, Y. Ji, M. Zhang, G. Fox, M. Marathe. **IrrMap: A Large-Scale Comprehensive Dataset for Irrigation Method Mapping**, *ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD 2025)*.

- Released a national-scale dataset and benchmark for irrigation mapping via remote sensing.
- Evaluated segmentation models for multimodal geospatial learning and irrigation classification.

N. C. Mandal*, O. B. Hoque*, A. Adiga, S. Swarup, S. K. Nouwakpo, A. Wilson, M. Marathe. **Knowledge-Guided Deep Learning for Advanced Irrigation Method Mapping (KIIM)**, *International Joint Conference on Artificial Intelligence (IJCAI 2025)*.

- Combined domain knowledge with multichannel sensing for knowledge-informed irrigation mapping.

- Used transfer learning to improve generalization with limited labeled data.

O. B. Hoque*, N. C. Mandal*, M. Wilson, S. Swarup, M. Marathe, A. Adiga. **PRISM-CAFO: Prior-Conditioned Remote-Sensing Infrastructure Segmentation and Mapping for CAFOs**, *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV 2026)*.

- Applied DINOv2 and SAM2 for multimodal detection and segmentation of CAFOs.
- Integrated priors and attention for interpretable, cross-modal infrastructure classification.

N. C. Mandal, G. Uddin. **An Empirical Study of IoT Security Aspects at Sentence-Level in Developer Textual Discussions**, *Information and Software Technology (Elsevier, Volume 150) (Top Journal in Software Engineering)*.

- Used LLMs like BERT and GPT to classify IoT security topics in text.
- Developed Stack Overflow tools to flag and summarize security-related discussions.

G. M. Shibli, M. T. R. Shawon, A. H. Nibir, M. Z. Miandad, N. C. Mandal. **Automatic Back Transliteration of Romanized Bengali (Banglish) to Bengali**, *Iran Journal of Computer Science (Springer)*.

- Built a deep learning model with GPT-3 embeddings for Banglish-to-Bengali transliteration.
- Deployed a real-time tool for multilingual translation and script correction.

NOTABLE RESEARCH PROJECTS

Epidemic Forecasting: Built OptimWISE, an adaptive ensemble that optimizes probabilistic forecasts via Weighted Interval Score (WIS). **Operational in CDCs FluSight Hub** for real-time influenza hospitalization forecasting.

Node Classification in Textual Graphs: Designed a hybrid GNNGAT model enriched with LLM-derived text embeddings to jointly learn graph structure and semantic signals, improving classification performance on text-attributed graph benchmarks.

Fallowing Land Identification: Developing a diffusion-based generative model that forecasts next-season fallowing probability, enabling drought and groundwater stress-testing scenarios. (Ongoing)

LEADERSHIP, MENTORSHIP & PROFESSIONAL SERVICE

- **Lead Organizer**, UVA Computer Science Research Symposium 2025 coordinated panels, speakers, and logistics.
- **Mentored two undergraduate** researchers on deep learning and multimodal geospatial AI projects (2025).
- Reviewer for KDD 2025 and KDD 2024 top-tier international conferences on Data Mining and AI.
- **Leadership Chair**, UVA CSGSG-2025 representing graduate student interests in departmental research.
- Organizing Committee Member, International Conference on Computer and Information Technology, 2020.

FELLOWSHIPS, HONORS & AWARDS

- Ph.D. Fellowship from the Department of Computer Science, University of Virginia.
- Travel Grants NeurIPS 2025 and IJCAI 2025.
- **Ranked 1st** in section merit list (among 60 students), BUET.
- Deans List of Honor (based on B.Sc. result) and University Merit List, BUET 2017 to 2019.

PROGRAMMING SKILLS

Languages: Python , Java , C , C++ **Technologies:** NumPy, SciPy, Scikit-Learn, TensorFlow, Keras, PyTorch, PyTorch Lightning, CUDA, Distributed Training (DDP) , OpenCV, HuggingFace Transformers, SentenceTransformers, OpenAI GPT-3/ChatGPT API, LoRA/QLoRA (PEFT), NLTK, Gensim, FastText , Rasterio, GDAL, GeoPandas, QGIS, Google Earth Engine, AlphaEarth Embeddings, Earth Explorer , Git, Bash, Slurm, Weights & Biases, Optuna, Docker, Linux